ADARE: Learning-Guided Adaptive Control for Many-Objective Workflow Scheduling in Heterogeneous Edge/Fog/Cloud Systems

Abstract. ADARE (Adaptive Data-driven Algorithm for Resource Evolution) is a learning-guided adaptive control layer for many-objective workflow scheduling in heterogeneous Edge/Fog/Cloud systems. Instead of changing the Non-dominated Sorting Genetic Algorithm III (NSGA-III) backbone, ADARE controls operator choice online with a non-stationary multi-armed bandit policy using Upper Confidence Bound (UCB), then couples this control with density-aware mutation and archive-guided refinement. This design keeps the optimization process interpretable while adapting to phase changes in the search landscape. We evaluate ADARE on Montage_25, CyberShake_30, and Epigenomics_24 with a strict fairness protocol: same evaluator, same budget, same seeds, and paired 20-run comparisons per benchmark. Across the final protocol, ADARE wins 18/18 quality comparisons and 15/15 core-quality comparisons (Hypervolume (HV), Inverted Generational Distance (IGD), spacing, epsilon, coverage). Mean gains are +26.80% on core quality, +10.70% on best-objective metrics, and +5.50% on runtime versus NSGA-III. In practical terms, the method improves both front quality and objective extremes without requiring heavy offline training. Runtime benefits are benchmark-dependent, with the smallest effect on Epigenomics, but still positive in the final 20-run setup. ADARE therefore provides a robust and reusable online control mechanism for evolutionary scheduling under heterogeneous compute, bandwidth, cost, and energy constraints.

Keywords: Online learning · Many-objective optimization · NSGA-III · Multi-armed bandits · Edge/Fog/Cloud · Workflow scheduling

1 Introduction

Modern distributed applications increasingly execute over heterogeneous computing strata, where edge nodes, fog servers, and cloud resources expose markedly different latency, bandwidth, energy, and cost profiles. In this context, workflow scheduling is not only a resource allocation problem; it is also a *control problem under uncertainty* over a search landscape that changes as candidate schedules evolve. We therefore study ADARE (Adaptive Data-driven Algorithm for Resource Evolution) as an online control layer embedded in a many-objective evolutionary loop, rather than as a stand-alone scheduler replacement.

NSGA-III remains a strong baseline for many-objective optimization because of its reference-point selection mechanism [7]. However, workflow-scheduling implementations commonly keep crossover and mutation probabilities fixed across the entire run. In heterogeneous Edge/Fog/Cloud settings, this assumption is fragile: an operator that is useful for early exploration can later damage partial structures once the population starts refining narrow trade-off regions. In practice, this can reduce either convergence pressure or diversity exactly when the search distribution starts to shift.

In our early paired runs, this phase-dependent behavior appeared repeatedly in the convergence traces before it became obvious in the summary tables. That pushed us away from redesigning the whole optimizer and toward a more targeted intervention: keep the NSGA-III selection backbone, but let a small online controller adapt crossover choice from observed Pareto-quality rewards. This framing also follows a learning-driven decision view: the optimization engine acts as the environment, operator choice is the action, and search feedback becomes the training signal.

Contributions. The four contributions below came out of repeated implementation iterations in our paired runs and reruns.

1. First, we cast crossover control in ADARE as a *non-stationary multi-armed bandit (MAB)* problem and implement an online UCB controller whose rewards are updated from a Pareto-quality proxy.
2. Second, we add density-aware mutation and archive-guided refinement, while keeping NSGA-III selection fixed for fair comparison.
3. Third, we report a statistically grounded evaluation on three workflow instances with **20 paired runs** per benchmark, and we use Wilcoxon signed-rank tests together with Cliff's delta effect sizes.
4. Finally, we provide computational complexity analysis and runtime profiling; the extra control layer remains modest relative to schedule evaluation cost.

The remainder of this paper is structured as follows: Section 2 reviews related work, Section 3 presents the problem modeling, Section 4 describes the proposed approach, and Section 5 details the experimental protocol and experimental results. Section 6 discusses the findings and their limitations, and Section 7 concludes the paper and outlines future research directions.

2 Related Work

2.1 Workflow Scheduling in Edge/Fog/Cloud Systems

Workflow scheduling in heterogeneous distributed systems is classically formulated as a Directed Acyclic Graph (DAG) mapping problem with precedence and communication constraints. Heuristic methods such as Heterogeneous Earliest Finish Time (HEFT) remain competitive for makespan-focused scheduling because of their low computational cost [1], yet they are not naturally designed

for multi-objective trade-offs involving latency, energy, and monetary cost. The fog computing paradigm further amplifies this challenge by introducing an intermediate layer with distinct communication and compute properties [3].

For this reason, many-objective formulations are more appropriate in heterogeneous edge, fog, and cloud orchestration than single-score scheduling. The scheduler should expose trade-off solutions rather than a single operating point. In the workflow community, benchmark instances such as Montage, CyberShake, and Epigenomics are widely used to stress both compute and communication behavior; we rely on public workflow repositories and Workflow Commons (WfCommons)-style instances for reproducible experimentation [6].

2.2 Baseline Many-Objective Evolutionary Algorithms

With a moderate objective count, many studies still start from Non-dominated Sorting Genetic Algorithm II (NSGA-II) [4]. In our comparisons, NSGA-III keeps the same lineage and uses reference points to retain selection pressure in many-objective settings [7]. Multiobjective Evolutionary Algorithm based on Decomposition (MOEA/D) [5] is different. It decomposes the problem into coordinated scalar subproblems. We keep this trio because it exposes distinct search biases during runs: dominance ranking, reference-point niching, and decomposition.

In practice, however, workflow scheduling studies often keep crossover and mutation schedules fixed because dynamic operator policies are harder to instrument and compare reproducibly. That engineering convenience becomes a methodological limitation in our setting. ADARE keeps the many-objective selection backbone intact (NSGA-III in the current artifact) and changes the control policy around operator choice and mutation behavior as the search state evolves.

2.3 Learning-Guided Adaptive MOEAs

Adaptive operator selection and reinforcement-learning-assisted evolutionary algorithms have become increasingly relevant in recent years. OVEA introduces Q-learning-driven operator adaptation in a reference-vector-based Multiobjective Evolutionary Algorithm (MOEA) [10]. A recent survey by Song et al. synthesizes the broader design space of reinforcement learning for evolutionary algorithms and highlights online control as a key direction for scalable adaptive search [11]. More recently, Wang et al. propose an MRL-MOEA framework based on multi-role learning in a many-objective setting [12], and Xue et al. report a Q-learning-assisted MOEA/D variant with adaptive weight adjustment [13].

These studies support the broader premise of *learning-guided evolutionary control*. In our setting, the key distinction is not only whether learning is used, but where it is inserted in the optimization loop and what runtime cost it introduces. Most published results target generic benchmark suites rather than workflow scheduling with heterogeneous costs and bandwidth asymmetry. ADARE addresses that systems-specific gap while keeping a controller that can be profiled directly on a local workstation.

Table 1. Compact Comparative Landscape (Baselines and Adaptive MOEA Positioning)

Method / family	Many-obj.	Online learning	Adaptivity	Relevance to this study
NSGA-II [4]	Med.	No	No	Standard baseline for multi-objective optimization; included for methodological positioning.
NSGA-III [7]	High	No	No	Direct paired baseline in the artifact; strong selection, though operator schedule remains static.
MOEA/D [5]	High	No	No	Strong decomposition baseline; included for comparative positioning.
Recent adaptive MOEAs [10,12,13]	High	Yes	Yes	Learning-guided control in generic MOEA settings; limited workflow-specific evidence.
ADARE (this work)	High	Yes (UCB)	Yes	Workflow-specific learning-guided adaptive control with fairness-controlled evaluation.

2.4 Comparative Landscape of Cited Baselines

Table 1 summarizes the roles of the most relevant baselines and adaptive MOEA families discussed above. The table is intentionally split between *algorithmic positioning* and *artifact availability*: in the current code artifact, NSGA-III is the implemented baseline for direct paired comparison, while NSGA-II and MOEA/D are included as methodological baselines for broader positioning.

With that positioning in place, we can state the scheduling model and the objective vector actually used by the online control loop.

3 Problem Modeling

3.1 System Model

We consider a workflow represented as a directed acyclic graph (DAG) $G = (V, E)$, where each task $v_i \in V$ carries a computational load and a data transfer footprint, and each directed edge encodes a precedence relation with communication dependency. The execution platform is a heterogeneous resource set R spanning edge, fog, and cloud nodes, each characterized by processing rate, communication bandwidth, monetary processing cost, and power usage.

A candidate schedule is encoded as a task-to-node assignment vector

$$x = (x_1, \ldots, x_{|V|}), \qquad x_i \in \{0, 1, \ldots, |R| - 1\}. \tag{1}$$

Given a topological order of the DAG, the evaluator simulates task start and finish times under node availability and inter-node transfer delays.

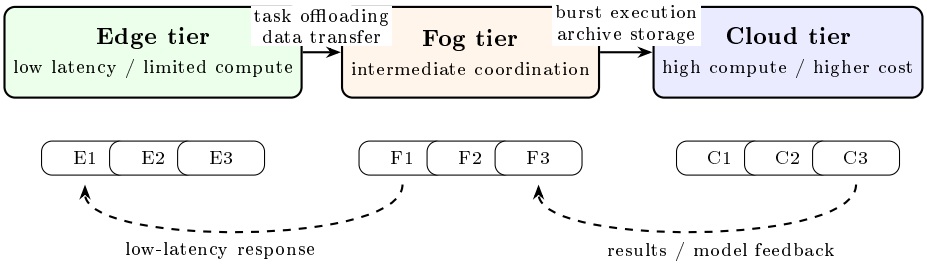


Fig. 1. Simplified Edge/Fog/Cloud execution architecture used to motivate the scheduling model. The heterogeneous compute, bandwidth, and cost properties across tiers create the many-objective trade-offs optimized by ADARE and NSGA-III.

Figure 1 clarifies the systems setting: edge nodes favor low-latency execution, fog nodes provide coordination and intermediate capacity, and cloud nodes provide stronger compute at a different cost/bandwidth profile. This heterogeneity is the reason a static operator schedule can become suboptimal during search.

3.2 Objective Vector

ADARE and NSGA-III optimize the same four-objective minimization vector:

$$\min \ \mathbf{f}(x) = \left(f_{\mathrm{mk}}(x), f_{\mathrm{lat}}(x), f_{\mathrm{cost}}(x), f_{\mathrm{eng}}(x)\right), \tag{2}$$

where:

- f_{mk} is the workflow makespan (maximum task completion time);
- f_{lat} aggregates task completion delays relative to readiness times;
- f_{cost} accumulates compute-time-weighted processing cost across assigned nodes;
- f_{eng} accumulates compute-time-weighted working power as an execution energy proxy.

Let C_i, R_i, I_i, ρ_{x_i}, c_{x_i}, and $p^{\mathrm{wrk}}_{x_i}$ denote completion time, readiness time, task instructions, node processing rate, node cost, and node working power; cost and energy use execution-time weighting I_i/ρ_{x_i}. The evaluator computes these metrics as:

$$\begin{aligned} f_{\mathrm{mk}}(x) &= \max_{i \in V} C_i, \\ f_{\mathrm{lat}}(x) &= \sum_{i \in V} \left(C_i - R_i\right), \\ f_{\mathrm{cost}}(x) &= \sum_{i \in V} \left(\frac{I_i}{\rho_{x_i}}\right) c_{x_i}, \\ f_{\mathrm{eng}}(x) &= \sum_{i \in V} \left(\frac{I_i}{\rho_{x_i}}\right) p^{\mathrm{wrk}}_{x_i}. \end{aligned} \tag{3}$$

This reflects the core edge/fog/cloud tension: faster assignments can reduce completion time while increasing cost or energy, so we target a diverse Pareto set rather than a single operating point.

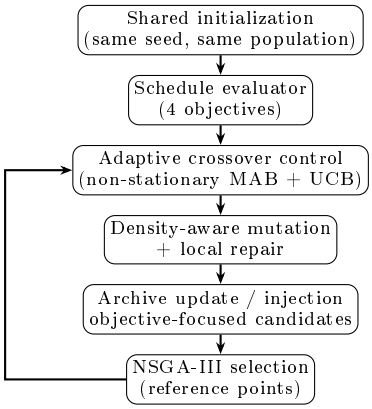


Fig. 2. ADARE as a learning-guided control layer embedded in the NSGA-III search loop.

3.3 Evaluation Targets and Quality Indicators

In addition to objective values, we evaluate search quality using hypervolume (HV), inverted generational distance (IGD), spacing, additive epsilon, and coverage. No single indicator was reliable enough on its own in our runs, so we keep this set to read front quality from complementary angles (dominance coverage, convergence to a reference front, and spread regularity). For ADARE, that distinction matters: the controller is meant to improve search behavior over time, not just one objective extreme.

With the objective vector and evaluation targets fixed, we now describe how the search loop uses this feedback online.

4 ADARE Methodology: Learning-Guided Adaptive Control

4.1 Design Overview

ADARE keeps the many-objective selection backbone of NSGA-III and replaces fixed operator control with an online adaptive layer. Figure 2 summarizes this organization.

We kept this architecture on purpose: replacing the full optimizer with a monolithic learner would blur attribution in paired comparisons. Instead, we adapt operator choice and diversity recovery while keeping NSGA-III selection unchanged, which preserves interpretability and a practical runtime profile.

4.2 Learning Formalization: Non-stationary MAB for Operator Control

At each generation, ADARE selects a crossover operator from a small portfolio (one-point, two-point, and uniform variants). We model this choice as a

multi-armed bandit (MAB) problem. The formulation is intentionally local: a compact controller is easier to instrument, debug, and compare fairly than a higher-capacity policy module in this artifact setting. The reward distribution is non-stationary because population diversity, front geometry, and feasibility structure evolve during search [8,9], which makes static crossover probabilities a poor fit for heterogeneous workflow scheduling.

Let Q_k be the current quality estimate for operator k, and N_k its usage count. At decision step t, ADARE chooses the operator with the UCB rule:

$$a_t = \arg\max_k \left(Q_k + \sqrt{\frac{2 \ln t}{N_k + 1}} \right). \tag{4}$$

The second term maintains exploration, while the first term exploits operators with strong recent rewards.

Reward definition. We define the reward as a scalar Pareto-quality improvement proxy computed on the parent-offspring comparison:

$$r_t = \phi(\min\{p_1, p_2\}) - \phi(\min\{c_1, c_2\}), \tag{5}$$

where $\phi(\cdot)$ is a log-scaled scalarization proxy (smaller is better). We retained this proxy after a few implementation attempts because front-level indicators at every mating event made operator-level attribution both expensive and noisy. A positive reward indicates that the selected operator produced better offspring according to the proxy. ADARE updates the selected operator value by exponential smoothing:

$$Q_k \leftarrow (1 - \alpha) Q_k + \alpha r_t. \tag{6}$$

This keeps operator control in an online-learning regime, while the update itself remains cheap enough for clean overhead measurements. Algorithm 1 gives the exact decision/update sequence used per mating event.

Algorithm 1 ADARE UCB-Based Crossover Control

Require: Parent pair (p_1, p_2), scores Q, usage counts N, step t
1: Select operator index k with Eq. (4)
2: Apply crossover operator k to obtain (c_1, c_2)
3: Compute reward r_t with Eq. (5)
4: Update $Q_k \leftarrow (1 - \alpha) Q_k + \alpha r_t$ and $N_k \leftarrow N_k + 1$
5: **return** (c_1, c_2)

4.3 Density-Aware Mutation and Local Repair

ADARE's mutation layer adapts perturbation effort using generation progress and population diversity. When diversity collapses, mutation pressure increases; when diversity is healthy, mutation stays conservative. A heuristic task-node

ranking (processing speed, bandwidth, cost, power) biases part of the mutations toward plausible assignments, with a random fallback to avoid early overcommitment. Algorithm 2 summarizes the mutation and repair sequence used in each generation.

Algorithm 2 Density-Aware Mutation with Heuristic Guidance

Require: Individual x, diversity score d, generation g, task-node rankings $\mathcal{R}$
Compute mutation budget $B \leftarrow f(g, d)$ (larger when diversity decreases)
for $b = 1$ to B **do**
    Sample a task index i
    Set x_i using heuristic-ranked nodes or a random fallback
end for
Apply a low-probability gene-level mutation pass
Optionally perform a small greedy local repair
return mutated individual x

4.4 Archive Guidance and Objective-Focused Final Candidate Pool

ADARE maintains a bounded archive of strong non-dominated or high-value candidates. During evolution, archive injection is rate-limited to guide search without collapsing diversity. At the end of the run, ADARE separates two roles: (i) the final non-dominated population for Pareto-quality metrics (HV, IGD, spacing, epsilon, coverage), and (ii) an objective-focused pool augmented with archive and targeted candidates for best-objective reporting (e.g., makespan or latency extremes). Algorithm 3 details this construction.

Algorithm 3 Archive-Guided Final Objective Pool Construction

Require: Final population P, archive A, targeted candidate set K
$P^{\text{obj}} \leftarrow P \cup A$
for all $k \in K$ **do**
    if k improves at least one tracked objective extreme **then**
        Insert k into P^{obj}
    end if
end for
return P for Pareto-quality metrics and P^{obj} for objective-best metrics

4.5 Complexity and Measured Control Overhead

The schedule evaluator dominates runtime. For population size P, generation count G, task count $|V|$, and average dependency-processing factor $\bar{d}$, evaluation cost is approximately

$$\mathcal{O}(G \cdot P \cdot |V| \cdot \bar{d}). \tag{7}$$

The UCB controller adds constant-time operations per operator decision (mean update, counter update, logarithm, square root), while archive operations are bounded on a small structure. In this artifact, the schedule evaluator still dominates wall-clock time for realistic workflow sizes.

Table 2. Local Host Machine (from system diagnostics)

Item	Specification
System model	ASUS TUF Gaming A15 FA506NF_FA506NF (ASUSTeK COMPUTER INC.)
Host Operating System (OS)	Windows 11 Home 64-bit, Build 26200 (dxdiag timestamp: 2025-12-18)
Central Processing Unit (CPU)	AMD Ryzen 5 7535HS with Radeon Graphics, 12 logical CPUs, 3.300 GHz nominal
Random Access Memory (RAM)	16.000 GB installed (15.690 GB available in dxdiag)
Graphics Processing Unit(s) (GPU)	AMD Radeon(TM) Graphics + NVIDIA GeForce RTX 2050 (3.964 GB dedicated reported)
Display / mode	1920×1080; host reports 144 Hz internal panel and 75 Hz external display

To verify this empirically, we instrumented UCB selection and update, together with archive routines, on 15 representative runs (5 seeds per benchmark). The measured upper-bound control ratio is **5.17% on average** (maximum **5.34%**) of total ADARE runtime. The instrumentation was intentionally conservative at Python level, so the ratio should be read as an upper bound rather than an exact decomposition of every micro-operation. It nevertheless supports the intended near-5% overhead target while preserving measurable quality gains.

5 Experimental Protocol and Results

5.1 Artifact, Local Execution Context, and Hardware Configuration

All experiments were executed locally using the code in the current ADARE artifact (no cluster). We report the host details from system diagnostics because runtime differences at this scale are sensitive to workstation-level CPU and memory behavior.

Table 2 is operationally important because runtime gains in Python MOEAs depend on CPU generation and memory behavior. This became especially relevant when some longer runs were split into batches for execution stability. The local software stack used for the experiments is Python 3.13.1 with `numpy 2.2.5`, `matplotlib 3.10.8`, `deap 1.4`, `scipy 1.16.3`, and `pymoo 0.6.1.6`.

5.2 Benchmarks and Heterogeneous Resource Configuration

We evaluate three workflow instances commonly used in scientific workflow scheduling: `Montage_25`, `CyberShake_30`, and `Epigenomics_24`. These workflow families are available through public repositories used in the Pegasus and WfCommons ecosystems [6].

The heterogeneous resource model is defined in the experiment configuration. The tier template is aligned with the Edge-Fog Cloud reference setting

Table 4. 20-Run Summary (ADARE vs NSGA-III, means; NSGA-III in parentheses)

Benchmark	HV	IGD	Δ Mk. (%)	Δ Lat. (%)	Δt (%)
Montage_25	0.9493 (0.9232)	0.0600 (0.0624)	19.01	11.77	4.89
CyberShake_30	0.9645 (0.9303)	0.0633 (0.0718)	5.17	8.01	7.47
Epigenomics_24	1.0453 (1.0337)	0.0342 (0.0414)	1.18	4.34	4.13

proposed by Mohan and Kangasharju [2]. Table 3 reproduces the exact tier-level parameters used in the artifact.

Table 3. Resource Configuration Used in Experiments

Tier	Devices	Proc. rate	Cost	Idle P.	Work P.	Up/Down BW
Edge	5	1000	0.02	30	700	20480 / 20480
Fog	5	1300	0.48	30	700	10000 / 10000
Cloud	5	1600	0.96	1332	1648	100 / 10000

Devices: number of nodes; Proc. rate: processing rate in million instructions per second (MIPS); Cost: processing cost per second; Idle P./Work P.: power in watts; Up/Down BW: uplink/downlink bandwidth in megabits per second.

This table creates the optimization tension used throughout the study. In early debugging runs, it was easy to over-focus on cloud processing rate alone; the bandwidth asymmetry and cost/power gradients explain why that intuition is often wrong.

5.3 Fairness Protocol, Metrics, and Statistical Testing

ADARE and NSGA-III share the same evaluation function, generation budget, and initial population at each paired run. The artifact generates a shared initial population from the same seed before instantiating both algorithms, which enforces a fair comparison baseline.

We report **20 paired runs per benchmark**. In our runs on the workstation, benchmarks are executed sequentially (one workflow family at a time) to avoid runtime perturbations from concurrent processes. The statistical design is unchanged: paired 20-run comparisons.

Reported outputs include: (i) Pareto-quality metrics (HV, IGD, spacing, epsilon, coverage), (ii) runtime, and (iii) best objective values for makespan, latency, cost, and energy. For statistical validation, we use the paired Wilcoxon signed-rank test and Cliff's delta effect size [16,17].

5.4 Main Results: 20-Run Paired Comparison (ADARE vs NSGA-III)

Table 4 summarizes the main comparative outcomes (means over 20 paired runs). NSGA-III means are shown in parentheses for direct readability.

ADARE improves HV on all three workflows and improves makespan/latency extremes in every case. In the final 20-run configuration, IGD also improves on

Table 5. Statistical Evidence (20 Paired Runs): Wilcoxon p and Cliff's δ

Benchmark	Metric	Wilcoxon p	Cliff's δ
Montage_25	HV	1.05×10^{-4}	0.645 (large)
	Coverage	1.33×10^{-2}	0.440 (medium)
	Runtime	6.04×10^{-3}	0.725 (large)
CyberShake_30	HV	1.91×10^{-6}	0.980 (large)
	IGD	9.54×10^{-7}	0.940 (large)
	Runtime	1.97×10^{-4}	0.760 (large)
Epigenomics_24	HV	9.54×10^{-7}	0.960 (large)
	Spacing	9.54×10^{-7}	0.965 (large)
	Runtime	2.66×10^{-2}	0.325 (small)

all three workflows. This is useful because it means the stronger extremes and spread do not come at the expense of distance-to-reference quality in the reported protocol.

Table 5 confirms that the quality gains are not only visual. HV improvements are significant on all three workflows, and the selected runtime comparisons are also significant in this final protocol. Epigenomics nevertheless remains the benchmark where runtime gains are the smallest in effect size (small Cliff's δ), even when the direction is favorable.

In our runs across the three benchmarks, ADARE wins **18/18** quality comparisons (including runtime) and **15/15** core-quality comparisons (excluding runtime). Average gains are **26.80%** in core quality, **10.70%** in best-objective metrics, and **5.50%** in runtime.

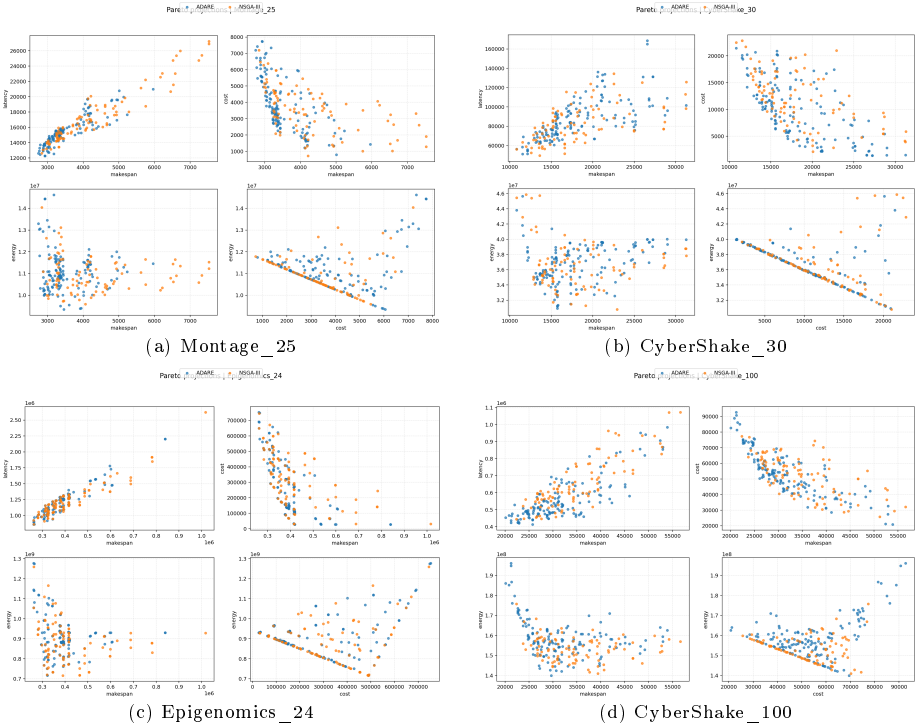
5.5 Ablation Protocol (V1–V5) and Artifact-Ready Variant Definition

To isolate module effects in our runs, we keep the same paired-run fairness constraints and grow the ablation step by step. We start with **V1** (NSGA-III baseline), then add UCB crossover control in **V2**, density-aware mutation in **V3**, partial archive injection/final-pool guidance in **V4**, and full ADARE in **V5**.

Table 6. Numerical Ablation Results (V1–V5, 20 paired runs per benchmark)

Variant	UCB	Mut.	Arch.	Core (%)	Obj. (%)	Time (%)	Wins
V1	N	N	N	0.00	0.00	0.00	–
V2	Y	N	N	1.03	6.46	7.76	8/15
V3	Y	Y	N	1.95	7.29	8.63	11/15
V4	Y	Y	P	17.37	7.54	8.06	15/15
V5	Y	Y	Y	30.60	9.47	6.93	15/15

Numerically, V2 and V3 improve over V1 but remain less consistent (quality wins: 11/18 and 14/18), while V4 and V5 are both stronger (18/18), with V5 giving the best mean core gain. In Table 6, N/Y/P denote No/Yes/Partial. For a metric m, we compute $g_m(\%) = \frac{A_m - N_m}{|N_m|} \times 100$ for max metrics and $g_m(\%) = \frac{N_m - A_m}{|N_m|} \times 100$ for min metrics. Core (%) averages g_m over HV, IGD, spacing,



(a) Montage_25 (b) CyberShake_30

(c) Epigenomics_24 (d) CyberShake_100

Fig. 3. Representative Pareto projections (ADARE vs NSGA-III) across three main workflow families, plus a larger available case (CyberShake_100) from an exploratory paired run set.

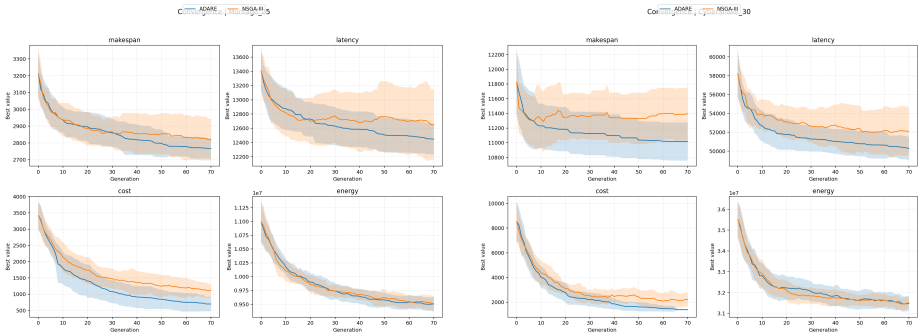
epsilon, and coverage; Obj. (%) averages g_m over makespan_best, latency_best, cost_best, and energy_best; Time (%) is runtime gain; Wins counts positive core gains across the three benchmarks (out of 15). Positive values indicate ADARE gains.

5.6 Qualitative Pareto and Convergence Analysis

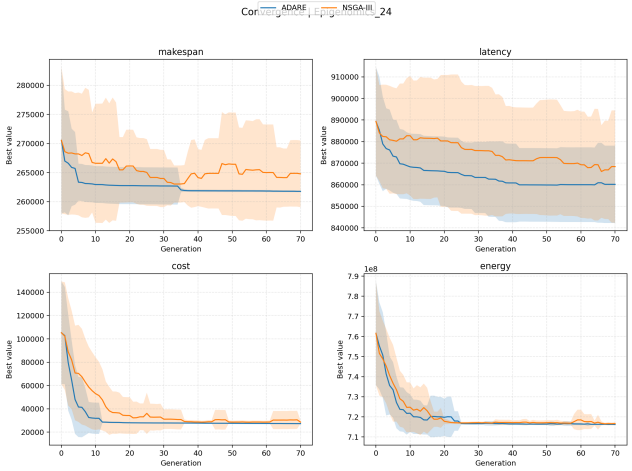
We inspect representative Pareto projections and convergence traces next because the tables alone do not show *how* the search behavior changes.

Figure 3 is consistent with the quantitative results: ADARE extends favorable regions on Montage, broadens coverage on CyberShake, and improves spread on Epigenomics while preserving strong objective extremes. The larger CyberShake_100 projection remains coherent with that trend and gives an additional visual stress case beyond the three main benchmarks.

Figure 4 makes the control effect visible in our runs. This matters. We see earlier trajectory separation on Montage and CyberShake, and later generations stabilize differently. Epigenomics still improves, mainly through front structure, while runtime movement stays smaller.



(a) Montage_25 (b) CyberShake_30



(c) Epigenomics_24

Fig. 4. Convergence traces (best objective values by generation) for Montage, CyberShake, and Epigenomics. Larger panels are used to improve readability of generation-wise trajectories and confidence bands.

At this point, the practical question is less "does ADARE ever help?" and more "when does the adaptive layer help most, and where are its limits?" The discussion below addresses that question directly.

6 Discussion

6.1 Why Static NSGA-III Underperforms in Heterogeneous Environments

In our setting, the main weakness of a static NSGA-III pipeline is not the selection operator itself; NSGA-III's reference-point mechanism remains a strong many-objective baseline. The recurring issue is *operator control rigidity*. Fixed crossover and mutation schedules assume a reward landscape that stays roughly stable, but workflow scheduling across heterogeneous edge, fog, and cloud resources shows phase changes during search.

Early generations benefit from broad exploration to discover useful task-node assignments under precedence and bandwidth constraints. Later generations benefit more from operators that preserve partial structure and refine local trade-offs. A static schedule cannot represent this shift explicitly, while ADARE's controller adapts operator preference from observed rewards. The convergence plots were useful here because they exposed this phase dependence before the aggregate statistics made it obvious.

6.2 Justification of Experimental Choices

Population size and generation count. We use population size 100 and 70 generations to keep comparisons computationally affordable on a local workstation while preserving enough diversity for four-objective optimization. Increasing population size improves front coverage but raises evaluator cost in practice through more offspring evaluations and more frequent archive maintenance. The chosen configuration is a compromise between statistical replication and per-run search depth.

Seeding and fairness. Seed control is not a cosmetic choice. In stochastic MOEAs, comparing unmatched runs can confound algorithmic differences with initialization variance. Our paired design uses the same initial population for ADARE and NSGA-III for every run, which makes Wilcoxon and Cliff's delta more informative.

Why 20 runs. The original artifact analysis used 6 runs. In our runs, that count exposed strong effects but remained fragile for smaller effect-size comparisons, especially runtime relative to HV/IGD/spacing gains. We moved to 20 paired runs to stabilize mean estimates and effect sizes. Table 5 shows that directions remain favorable while effect sizes still vary by benchmark and metric.

6.3 Sensitivity and Robustness Trends

Our observations suggest three sensitivity patterns.

1. **Run count sensitivity**: metrics with smaller effect sizes (most clearly runtime on Epigenomics in our final protocol) are sensitive to the number of repetitions and should not be interpreted from very small samples.
2. **Mutation/Archive sensitivity**: ADARE's strongest empirical gains are associated with spacing, coverage, and objective extremes, indicating that density-aware mutation and archive guidance dominate the observed improvements in difficult landscapes.
3. **Benchmark dependence**: ADARE delivers strong quality gains on all three workflows, but quality and runtime do not move with the same amplitude on every instance. Epigenomics shows a smaller runtime effect size than CyberShake or Montage despite clear quality improvements.

These patterns support the adaptive-control interpretation in our runs. Under the reported 20-run protocol, ADARE is superior on all tracked quality metrics, yet the *magnitude* of improvement still depends on benchmark structure. For that reason, we avoid benchmark-independent effect-size claims from smaller repetition counts, even when win/loss counts look unambiguous.

7 Conclusion

We reframed ADARE as an online control layer for many-objective workflow scheduling rather than as a simple scheduler variant. This framing is technically useful because it explains why operator-level learning is central to the method, while the evolutionary backbone can remain largely unchanged.

Under a fairness-controlled protocol (same initialization, same budget, same evaluator) and a stronger statistical setup (20 paired runs per benchmark), ADARE improves overall many-objective search quality and runtime against NSGA-III on three heterogeneous workflow instances. In our runs, gains are strongest in HV, coverage, spacing, and objective extremes, and CyberShake shows the clearest combined benefit. Runtime also improves on all three benchmarks in the final protocol, although the Epigenomics runtime effect remains smaller than its quality effects. Measured control overhead stays near 5% on average, preserving practical usability on a local workstation.

For the ECML PKDD framing, the main takeaway is that workflow scheduling in heterogeneous distributed systems benefits from treating evolutionary operator choice as an *online learning control problem*. Future work will extend this evaluation to larger workflow instances, finalize direct NSGA-II and MOEA/D implementations, and add direct comparisons with recent learning-based state-of-the-art methods such as QLMOEA/D-AWA under the same evaluator.

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